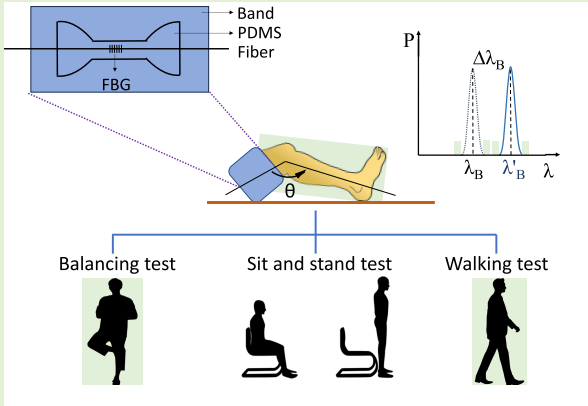
Anomaly Detection in Gait Cycles Using a PDMS Embedded FBG Sensor

Archana Thrikkaikuth Chalackal and Srijith Kanakambaran

***Abstract*—Walking is one of the most natural yet risky motor functions performed by the body. The manner of walking, or gait, depends on the proper functioning of the lower limbs, alignment of ankle and knee joints, and the balance of the body, without which our body is prone to falls. Aging leads to improper functioning of the lower limbs leading to abnormalities in gait and increased risk of falling in the elderly. Hence, gait analysis is necessary in providing better healthcare and fall prevention for elderly. In this work, a polydimethylsiloxane (PDMS)-embedded fiber Bragg grating (FBG)-based sensor in a wearable band is proposed for the detection of improper gait with focus on knee angle measurement. The PDMS-embedded sensor, in the dimensions of ASTM D412 standard, is placed on the thigh for experimental measurements. The sensor is found to exhibit a nonlinear response to bending and extension of the knee which fits a second degree polynomial implying the requirements for advanced methods for linearization to enable absolute measurement of angles. From the gait perspective, the change in Bragg wavelength during activities as measured in relation to a reference reading holds importance in the identification of anomalies. Hence, tests were conducted to monitor the shift in wavelength during activities like walking (normal and time controlled) to monitor the gait, balancing on one leg and sit and stand trials. The proposed sensor is capable of distinguishing the different activities and variations in gait on path with and without obstacles.**



***Index Terms*—Fiber Bragg grating (FBG), knee angle, optical fiber sensor, polydimethylsiloxane (PDMS), wearable sensor.**

I. Introduction

WALKING is a complex motor function which is performed automatically by adults which requires synchronized movement of each lower limb part. The study of human locomotion is referred to as gait analysis. The time period between two consecutive contact of the heel of the same foot with the floor constitutes a gait cycle. There are two phases in a gait cycle: stance phase—period of contact with the ground and swing phase—period of no contact with the ground. Stance phase lasts for 60% and swing phase for 30% of the gait cycle [1], [2].

The study of gait involves two parts—kinematic analysis which involves the study of joint angles, angular velocities and acceleration and kinetic analysis which involves the study of ground reaction and joint forces. Disorders in any part of the body, such as spinal cord issues, decreased knee flexion, issues with the alignment of the hip, ankle and knee joints, can alter the gait pattern of an individual [3], [4], [5]. Improper gait cycle can also make the body prone to falls. The ease of walking is different for younger and older adults. Elderly often find it difficult to walk with ease due to gait impairments making them prone to falls leading to serious injuries and psychosocial consequences [6]. Hence, analysis of clinically relevant gait parameters like knee angles in elderly is necessary for fall prediction and the early detection and treatment of gait disorders.

Angle measurements at knee and ankle are necessary for fall prevention. Most of these measurements need to be taken on complex joints and curved surfaces. Therefore, it is convenient to use a compact, wearable sensor which is of least disturbance to the subject for proper measurement. Several procedures also require the measurement of multiple parameters. Few

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This work involved human subjects or animals in its research. The authors confirm that all human/animal subject research procedures and protocols are exempt from review board approval.

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applications involve measurement in environments with high magnetic fields such as during magnetic resonance imaging. This limits the usage of conventional sensors which can be overcome using optical fiber-based sensors (OFSs). In such sensors, the light guided by an optical fiber is modulated by perturbations like strain and temperature. Numerous applications, including structural health monitoring, acoustic sensing, aerospace, chemical sensing, biomedical sensing, and more, use OFS sensors because of their advantages over conventional sensors, which include small size, high sensitivity, immunity to electromagnetic interference, wide dynamic range, ease of multiplexing, and the capacity to withstand harsh environments [7], [8].

The small size, stability, better production technology, tolerance to harsh environments, and low-loss of fiber Bragg gratings (FBGs) makes it the highest commercial OFS [9], [10]. FBGs have been widely used in the development of wearable sensors. Using this sensor, multiple parameter detection may be possible with the same fiber. Coating FBG-based sensors can increase their sensitivity. Polydimethylsiloxane (PDMS) is one such coating material that improves the sensor's performance and contributes to a more precise and consistent output. Since PDMS is safe, nontoxic, and biocompatible, it is a great option for coating or embedding FBGs utilized in biomedical sensing [11], [12]. Due to its flexibility, FBGs embedded in it have been used to develop a number of wearable sensors for biomedical applications [13], [14], [15]. In this work, we propose a PDMS-embedded FBG sensor integrated into a wearable Velcro band, enabling a wearable structure for gait analysis with a key focus on knee angle monitoring. To the best of our knowledge, there has been no work reported till date on the study of gait cycles specifically addressing anomaly detection in gait using an FBG embedded in PDMS. As such, the unique contributions of this work are as follows:

1) The study of variation in Bragg wavelength with knee angle for an FBG sensor embedded in PDMS matrix of ASTM D412 dimension.
2) The investigation on capability of the sensor to detect the different phases in gait and change in Bragg wavelength for different activities like sitting, standing and balancing on flat and nonflat surface.
3) The detection of anomalies in gait pattern when a subject treads a path with obstacles.

The rest of this article is organized as follows. Section II describes the FBG sensing principle and the relevant literature on knee angle measurement using FBG-based sensors. Section III provides a summary of the embedding procedure and the methodology for knee angle measurement. This is followed by the results and discussions in Section IV which describes the results from the experiments conducted for gait analysis. The conclusions of the work are summarized in Section V.

II. Background

A. Sensing Principle of FBG

Fiber Bragg grating refers to periodic variations in the refractive index of the core of an optical fiber. The sensing principle of FBG is shown in Fig. 1. When FBG is exposed to broadband light, a narrow band of wavelengths centered on a particular wavelength is reflected. This wavelength is called Bragg wavelength (λ_B). The Bragg wavelength of FBG is given by

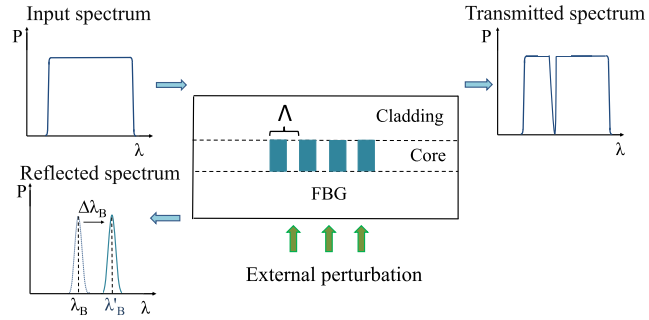

Fig. 1. Sensing principle of FBG.

$$\lambda_B = 2n_{\text{eff}}\Lambda \tag{1}$$

where n_{eff} is the effective refractive index of the fiber core and Λ is the grating period.

The presence of external perturbation such as temperature or strain leads to a change in n_{eff} and Λ causing a proportional shift in the Bragg wavelength. This enables the use of FBG as a sensor.

The shift in wavelength ($\Delta\lambda_B$) due to the effect of strain and temperature is given by

$$\frac{\Delta\lambda_B}{\lambda_B} = (1 - P_e)\epsilon + (\alpha + \zeta)\Delta T \tag{2}$$

where P_e, ϵ, α, ζ, and ΔT are the photoelastic constant, induced strain, thermal expansion coefficient, thermo-optic coefficient, and change in temperature, respectively [16].

Although bare FBG-based sensors perform well in various applications, addition of suitable coatings can improve their structure and sensitivity. A suitable polymer-based coating for FBG-based sensors is PDMS. It is flexible, biocompatible, and nontoxic and is suitable for development of wearable sensors for biomedical applications [17], [18].

B. Related Work

Knee flexion and extension are important parameters in gait analysis and provides information related to walking disorders and pathologies. This is also used to monitor the efficiency of therapies and in sports to improve the performance of athletes.

In [19], a sensor consisting of FBG encapsulated into a silicone brick attached to a knee brace was developed for static and dynamic monitoring of knee movements. Three-dimensional printed plates and screws holds the sensor to the knee brace. To study the influence of polymeric matrix length to sensor response, two sensors were developed-big (50 × 30 × 10 mm) and small (40 × 30 × 10 mm). The sensitivity was found to be 3.944 and 2.897 nm·mm^{-1} for the big and small sensors, respectively. The feasibility of the sensor was assessed by monitoring the knee flexion and extension during walking tests at different speeds. From the pattern of Bragg wavelength change, flexion and extension could be easily

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identified. The sensor demonstrated the capacity to track knee movements while walking and determine the number of stance and swing phases in gait at various speeds. Domingues et al. [20] proposed an accurate and noninvasive FBG-based sensor with kinesio tape to monitor the knee angular flexion/extension during gait. The Bragg wavelength shift was correlated with knee angle measurements from a goniometer attached to the leg. The knee flexion angles were increased at a rate of 5° up to 60° and then decreased to 0° at the same rate. Bragg wavelength shift was recorded for each step and showed exponential dependence with applied angle. The reliability of the sensor was tested by continuous monitoring of Bragg wavelength shift during individual normal walk of a female subject. The Bragg wavelength shift reflects angular movement of the knee as well as muscle activity (muscles tension) and different moments of the gait cycle. The miniature size of the developed sensor enables it to be integrated within a noninvasive wearable e-Health monitoring system. In [21], a wearable smart belt using an FBG sensor was developed for knee posture monitoring. The FBG was prestressed to about 2000 $\mu\epsilon$, embedded in a central slot of a silica gel base, and covered with a polyvinyl chloride (PVC) strip. Fabric belts were attached to secure the sensor on the knee. Knee posture changes caused wavelength shifts due to volume extension near the joint. Experiments involved bending the knee from 0° to 80° and measuring angles with a scale. The sensor has a measurement sensitivity of 0.05 nm/°. The belt was used to examine knee movements during static and jogging tests. The belt is also capable of monitoring movements at other joints like the waist, wrist, arm, and shoulders. A sensor configuration with FBG in PVC foil was proposed in [22] for knee joint angle measurements. The PVC foil is attached to an elastic knee band and pressure buttons were used as bonding elements. Walking and running treadmill tests at different speeds were conducted to evaluate the consistency and reliability of the system. The stance and swing phases were clearly distinguished from the results. Polylactic acid (PLA) rings with embedded FBG for monitoring elbow and knee joint postures were developed in [23]. The PLA rings were fabricated by 3-D printing at 220°C. The two smart rings installed at the elbow and knee joints had maximum measurement angles of 90° and 100° and measurement sensitivity of 5.6 pm/° and 27.6 pm/°, respectively. A wearable knee angle measurement device (KAMD) was proposed in [24], which employs a FBG sensor to transduce the angular movement between the shin and thigh into strain variations on a cantilever, which are then measured by the FBG sensor. The proposed device efficiently evaluates knee angles for both legs simultaneously in three sensitivity modes—low (0.012 nm/°), medium (0.023 nm/°), and high (0.033 nm/°), by varying distance between pin and rivet. The sensor offers a noninvasive and effective solution for gait analysis and rehabilitation monitoring. A wearable sensor utilizing an FBG sensor embedded in PDMS of dimensions 40 × 30 × 10 mm was designed in [14] for monitoring squat exercises. A linear relationship between knee angle and wavelength shift was obtained in a working range of 30–120° during knee flexion and extension. The wearable sensor successfully measured knee angle in real time and detected correct and wrong postures during the squat exercise. A wearable high-sensitive double-clad FBG sensor encapsulated in silica gel for monitoring body posture and movement was proposed in [25]. The sensors exhibited a sensitivity of 7.5 pm/° and 36 pm/° in elbow and knee joint static tests. The double-clad FBG sensors exhibited better stability and repeatability in comparison to single-mode FBG sensors. In [26], two wearable noninvasive devices, KAMD for dynamic knee angle measurement and forefoot load measurement device (FLMD) for dynamic measurement of load on the forefoot, were realized using FBG sensors. The KAMD transduces the angular movement between the thigh and the shin of a subject into strain variations over a cantilever, which is acquired by the FBG sensor bonded over it. The FLMD transduces the load variations on the forefoot of a subject to strain variations over a cantilever with FBG bonded on it. The FLMD in combination with KAMD provides an effective system for analyzing spatio-temporal parameters in gait.

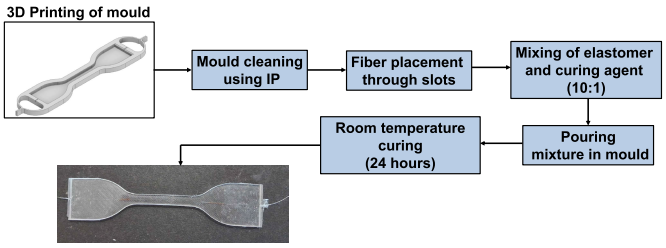

Fig. 2. Flowchart of embedding procedure.

In most studies mentioned above, the focus is on the detection of stance and swing phases in relation to the flexion and extension of knee at predetermined walking speeds. The gait pattern during normal speed of walking and the change in pattern in nonideal situations are not analyzed in detail. In this study, the authors attempt to acquire the gait patterns during natural walking and controlled walking, and detect anomalies in gait.

III. Methodology

A. Preparation of PDMS Embedded FBG Sensor

The sample under consideration is an FBG embedded using the flexible material PDMS (Sylgard 184, Dow Inc.) which has properties similar to elastomers and rubbers. The ASTM D412 standard for thermoplastic elastomers and vulcanized rubber samples is employed for mechanical characterization of PDMS. The geometric shape for the testing ensures that the elongation and the applied stress across the narrow region is uniform. Hence, the same dimensions were chosen as the shape of PDMS embedding for the sensing trials with the FBG-embedded halfway through the height of the matrix. A similar shape of the embedding has previously been reported in [27]. The mold design and the procedure for embedding FBG in PDMS is depicted in Fig. 2. A mold of ASTM D412 dimensions, length of 115 mm (gauge length of 25 mm), width of 6 mm, and thickness of 3 mm, was created in PLA using 3-D printing [28]. The mold incorporates side slots and a grasper to hold the fiber ends.

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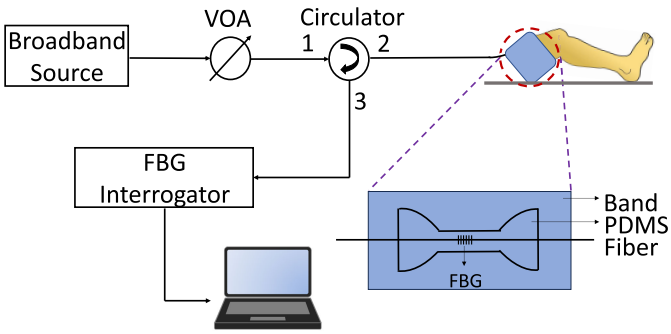

Fig. 3. Experimental setup for measurement of knee angles.

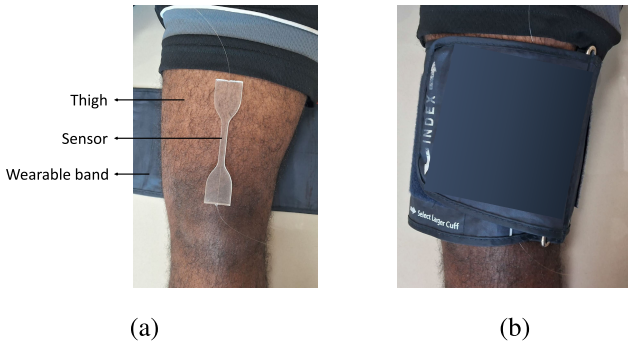

Fig. 4. Photograph of (a) sensor placement on the thigh and (b) Velcro band worn by the subject.

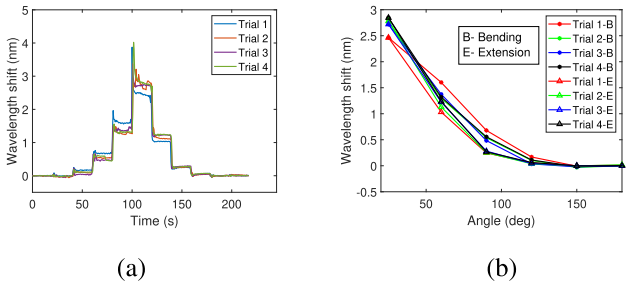

Fig. 5. Variation of $\Delta\lambda$ with (a) time and (b) knee angle during bending (B) and extension (E).

Isopropyl alcohol solution was used to clean the 3-D printed mold. The fiber to be embedded was inserted through the slots at the ends of the mold along the length. The depth of the slots was designed to ensure that the fiber is embedded at half the height and width of the PDMS embedding. The side slots and base edges of the mold were coated with petroleum jelly to prevent any PDMS leakage. Following this, the Sylgard 184 elastomer and its curing agent were mixed in the ratio of 10:1 and poured into the mold with fiber. The mixing ratio of 10:1 was chosen as it provided the best strain sensitivity as observed from the results of tensile testing [29]. The mixture was cured for 24 h at room temperature and the sample was removed.

B. Experimental Setup

The setup for measurement of the wavelength shift from the embedded FBG is shown in Fig. 3. A broadband light source (SLED broadband source, Denselight Semiconductors, DL-BX9-CS5153A) and a variable optical attenuator (VOA) are attached to port 1 of a circulator. The embedded FBG (Bragg wavelength 1551.2 nm and bandwidth 0.4 nm) is connected to port 2 of the circulator. An interrogator (Ibsen Photonics, IMON 512 USB Interrogator) is attached to port 3 of the circulator in order to track the Bragg wavelength of the FBG. During tests, the interrogator makes it possible to obtain the reflection spectrum and the time-varying Bragg wavelength. The FBG sensor embedded in a PDMS matrix of ASTM D412 dimensions with mixing ratio 10:1 was placed on the front of the thigh of the subject and the shift in wavelength was noted with time. The sensor was held using a wearable band with a Velcro to adjust the tightness. The major movements for knee joint are bending and extension. Each of these movements are facilitated by a specified muscle or a group of muscles on the upper and lower region of the knee. There are four extensors—quadriceps femoris, and seven flexors—hamstring, sartorius, gracilis, popliterus, and gastrocnemics, which directly affect the movement of the knee [30]. Among the above-mentioned flexors, hamstring muscles located at the back of the thigh are the primary knee flexors [31]. Quadriceps femoris located at the front of the thigh is the primary knee extensor. The vastus lateralis, vastus intermedius, vastus medialis, and rectus femoris are the four muscles that make up the quadriceps muscle, and their fibers make up the quadriceps tendon. At the knee, these muscles serve as primary extensors at the lower extremity of the knee [32].

Even though the muscles in the upper and lower knees contribute to knee movement, the sensor must be positioned in a convenient area to allow for proper sensor installation with reduced contact between the upper and lower legs. Hence, the experiments were conducted by placing the sensor on the front of the thigh. A mechanical goniometer was used as a reference for angle measurement.

IV. Results and Discussion

In this section, the results of the knee angle measurement using PDMS embedded FBGs are summarized followed by the angle monitoring during different activities.

A. Knee Angle Measurement

Initial experiments for knee angle measurement were performed using the PDMS embedded FBG sensor for a male subject (height: 161 cm, limb length: 44 cm) in the age range 20–25 years. Informed consent was obtained from the subject who volunteered for the study. The sensor placement using the Velcro band is shown in Fig. 4.

In all trials, the sensor was placed on the thigh such that the FBG embedded within the PDMS was positioned 8.5 cm above the patella. The tightness of the Velcro band was adjusted to ensure that the full range of motion was possible while preventing the sensor from slipping. The sensor was not permanently fixed on the band to facilitate the usage of the same sensor for further research on multiple joints and subjects. The Bragg wavelength shift was noted for change in angle from full extension to full bending in steps of 30°. The angles of bending were measured using a mechanical goniometer and the position of the foot corresponding to the

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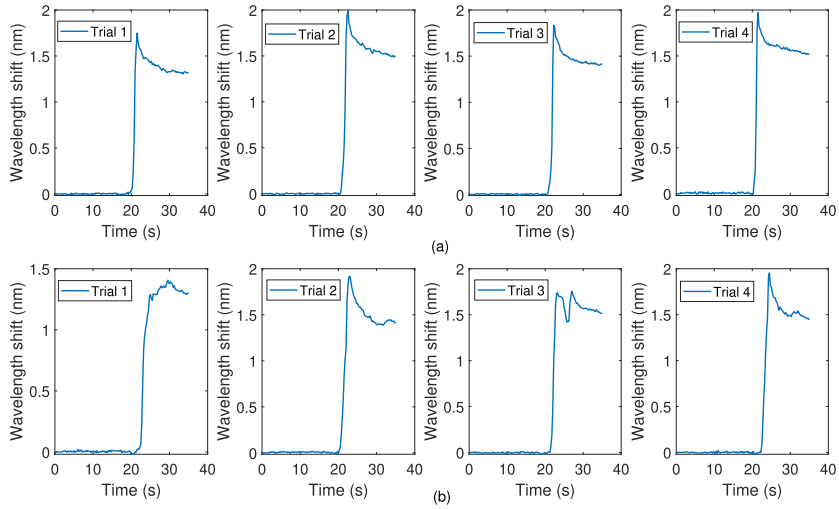

Fig. 6. Variation of $\Delta\lambda_B$ with time for four trials when subject balances on right leg on a (a) flat surface and (b) pillow.

bending at these angles were marked on the floor for reference. Each position was maintained for 20 s before changing the angle and the Bragg wavelength was recorded every 0.5 s by the interrogator. The plots of variation of Bragg wavelength with time and angle were obtained as shown in Fig. 5(a) and (b), respectively. The large change in λ_B for angle less than 100° can be due to the contact between the lower part of the knee and the thigh during bending.

The variation of wavelength with angle is obtained to be nonlinear. The mean variations in wavelengths for different bending angles can be approximated by second-degree polynomials $0.00017\theta^2$-$0.051\theta + 3.88$ and $0.0002\theta^2$-$0.057\theta + 3.94$ for bending and extension, respectively. The fit coefficients with standard deviations over four trials are 0.00017 ± 0.00005, -0.051 ± 0.006, and 3.88 ± 0.24 for bending and 0.0002 ± 0, -0.057 ± 0.004, and 3.94 ± 0.25 for extension, respectively. The coefficients of the fit polynomial show a close match, suggesting that the mean variations in Bragg wavelength during bending and extension are similar. A similar result has been observed by Domingues et al. [20] where the dependence of wavelength with angle is exponential.

The absolute measurement of angles require advanced methods for linearization of the response and look up table optimization [33], [34]. From the perspective of gait analysis, the relative changes in Bragg wavelength are more significant than the absolute change. Hence, the subsequent analysis focuses on the identification of anomalies in gait by studying the changes in Bragg wavelength during activities as measured in relation to a reference reading. The variation of Bragg wavelength during different gait-related activities involving the bending and extension of knee are explored in Section IV-B.

B. Knee Angle Monitoring During Activities

In addition to measurement of knee angles, Bragg wavelength shift was also observed when the subject performs different activities involving flexion and extension of the knee. These include balancing on a pillow, sit and stand cycles, and walking. The results from each of the activities with measurements performed with the sensor attached to the left leg are summarized in Sections IV-B.1–IV-B.4.

1) Balancing Test: The balancing test was conducted by making the subject stand and balance on one leg on a flat surface. In order to simulate the effect of loss of balance, the subject was made to balance on a pillow. The following series of actions were performed: standing at mean position for 20 s→step forward→balance on the leg for 10 s with data saved every 0.3 s. The results of the test when the subject balances on the right leg on a flat surface is shown in Fig. 6(a). The same test was conducted when the subject balances on a pillow and the corresponding variation in Bragg wavelength is shown in Fig. 6(b). When balancing on a flat surface, the Bragg wavelength shift is sharp and steep with a slight decrease following the peak. However, when balancing on the pillow, the variation is not sharp and the decrease following the sharp increase is not smooth.

The test was repeated when the subject balances on the left leg and the results for balancing on flat surface and pillow are shown in Fig. 7(a) and (b), respectively. In this case, for all the trials where the subject is standing on the pillow, an additional peak is observed, representing a slight loss of balance.

The variation in the waveforms were consistent across the four trials for the balancing trials on flat surface as well as pillow.

2) Sit and Stand Test: Following the balancing trials, the Bragg wavelength shift was analyzed during sitting and standing. Four trials were conducted with the repeated series of actions: sit for 20 s → stand for 20 s → sit for 20 s, and data were saved every 0.5 s by the interrogator. The variation in Bragg wavelength with time for the four trials is shown in Fig. 8.

The overall variation in $\Delta\lambda_B$ is observed to be similar for all the trials. The slight differences in the shifts may be attributed to the variation in tightness of the band and the position of the subject's legs while sitting for different trials.

3) Walking Test: The variation in Bragg wavelength during walking was analyzed when the subject moves at the natural walking speed. Four trials were conducted for a fixed length

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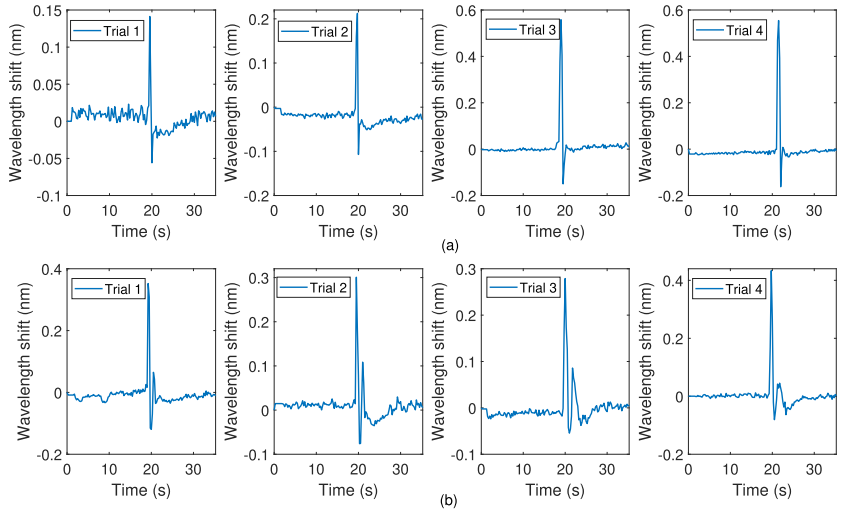

Fig. 7. Variation of $\Delta\lambda_B$ with time for four trials when subject balances on left leg on a (a) flat surface and (b) pillow.

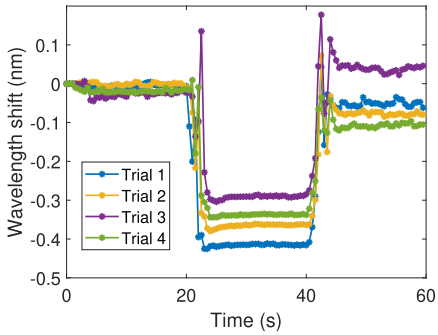

Fig. 8. Bragg wavelength shift with time for sit and stand trials.

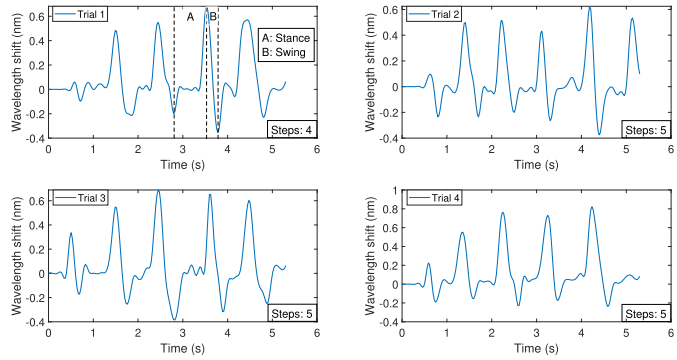

Fig. 9. Variation of Bragg wavelength with time for walking trials at normal pace after interpolation.

of the path with each trial taking around 5.5 s for completion of walking. The Bragg wavelength shift was saved every 0.1 s by the interrogator. The obtained waveforms have sharp transitions due to the longer time interval between data points. This can be overcome by reducing the time interval for recording data. However, due to the practical limitations of the interrogator, reducing the time interval for recording data can lead to data loss. Therefore, cubic spline interpolation was performed on the data to obtain smoother curves which are shown in Fig. 9. The number of steps taken by the subject during the trials was manually recorded by observation for reference and is mentioned in the figure. Each trial consisted of four to five steps. Step segmentation was performed manually by visually identifying the start and end of each gait cycle from the recorded sensor data as marked in trial 1 of Fig. 9. The average duration of a gait cycle was 0.99 s. The observed results for each step from the experiment have a close resemblance to the stance and swing phases recorded in [35] when an FBG is attached on the knee with K tape. This shows that the sensor can differentiate the phases of gait. The stance and swing phases recorded in the waveform are consistent with the number of steps taken by the subject. Further trials involved the subject walking on obstacles to record the difference in waveforms from normal walking.

4) Walking Test With Obstacles: Monitoring anomalies in gait cycles is significant in fall prevention and determining gait disorders. One approach to simulate the anomalies is to create a path with obstacles as shown in Fig. 10(a). Time limited walking trials were conducted to monitor the variation in wavelength with time for the subject during controlled walking with and without obstacles. The placement of the sensor on the knee is shown in Fig. 10(b).

The following were the conditions maintained to ensure uniform pace of walking across trials.

1) Each step was taken in an interval of 2 s.
2) The distance covered during every step is the same.

The sensor was fixed on the upper knee of left leg. Four trials each were conducted with the initial step taken with the left leg as well as the right leg. During the tests conducted for normal gait with data saved at 0.1 s intervals, data loss was observed in the saved data for a few trials. Hence, the subsequent trials were conducted with data saved at intervals of 0.3 s (the minimum time interval with no data loss).

The waveforms for trials were analyzed to observe changes in the results with and without obstacles. The variation in Bragg wavelength for trials where the subject places the left foot first and the right foot first are shown in Fig. 11(a) and (b), respectively.

In most trials, the peaks corresponding to the swing phase have a larger amplitude for trials with obstacles. This can be

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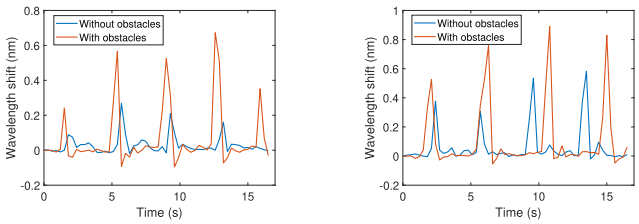
(a) (b)

Fig. 10. (a) Path with obstacles and (b) sensor placement during walking.



(a) (b)

Fig. 11. Bragg wavelength shift with time when subjects walk on path with and without obstacles for (a) left foot first and (b) right foot first.

due to the difficulty when the foot is placed on the ground with obstacles, leading to a larger bend in the knee when taking the next steps. The waveforms of trials with right foot first were similar to those observed in the trials with left foot first, with higher peaks for the swing phase for trials with obstacles. In all the trials with obstacles, distinct dips were observed at the beginning of the stance phase. The reason for this may be the uneven ground on which the foot is placed, requiring additional stress for firm placement, and due to the subject balancing on one leg for a short period of time. Unlike the larger dips observed during normal walk as shown in Fig. 9, these dips are smaller and consistent. This may be attributed to the controlled and slower nature of the walk.

A delay/advance in peaks is also observed in the trials. In the trials where the subject reported contact with sharper obstacles compared to the other trials, a large delay in peaks was observed toward the end of the trial. Delays and advances in steps are expected to occur when the subject either takes a slower or faster step to balance themselves on the path with obstacles. The loss of balance while taking steps is also clearly visible from the additional peak observed after stance, as marked in Fig. 12. This is similar to the waveform observed in the balancing test in Fig. 7(b). The results show that the sensor can be used to monitor the knee muscle movements during activities like sit and stand, balancing on one foot. The sensor is also capable of distinguishing normal and controlled walking and can also detect the changes in gait when obstacles are present in the path making it suitable for gait analysis. Additional experiments with the PDMS-embedded

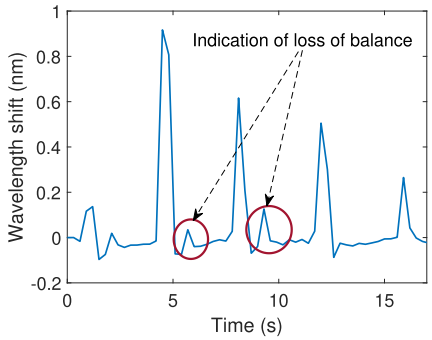


Fig. 12. Bragg wavelength shift with time for trial with maximum discomfort for the subject.

FBG samples are currently being conducted to evaluate their effectiveness in ankle and elbow angle monitoring and ground reaction force measurement.

V. Conclusion

This work explores the gait analysis with focus on knee angle measurement using a PDMS embedded FBG sensor in ASTM D412 dimensions. The variation of Bragg wavelength with angle for bending and extension were analyzed which follows a nonlinear variation. The mean variation in wavelengths for different bending angles is found to be approximated by second-degree polynomials $0.00017\theta^2$-0.051θ $+3.88$ and $0.0002\theta^2$-0.057θ $+3.94$ for bending and extension, respectively. Since the response is nonlinear, advanced methods for linearization of the response, look-up table optimization, are necessary to facilitate absolute measurement of knee angles. However, the change in Bragg wavelength during activities like walking, sitting and standing cycles, and balancing on flat and uneven surfaces have been assessed to determine the changes in gait. The sensor is capable of distinguishing different gait-related activities and can detect loss of balance while standing and walking. The variation in Bragg wavelength with time are consistent across multiple trials conducted for a particular activity. The sensor can also distinguish walking on path with and without obstacles. Hence, the wearable sensor is suitable for the assessment of gait-related activities and the subsequent determination of gait anomalies.

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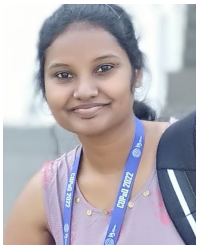

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